Task 1 Problem Definition and Business Understanding

SINTEF Ocean Real Vessel Data Challenge | TDT4259 Applied Data Science

25 October 2026 | Preliminary technical findings

1. Problem and practical objective

The objective is to estimate **vessel propeller shaft mechanical power (kW)** from operating and environmental measurements that do not originate from the same torquemeter channel. Such an estimate could provide a secondary indication of propulsion demand or help check whether a measured power value is plausible. Its usefulness for performance monitoring will depend on the predictive accuracy and the availability of input signals when an estimate is needed.

The task is formulated as **supervised regression**. The target is the recorded shaft mechanical power, and each training example represents an aligned one-minute period. The present formulation is a *completed-minute power estimate*: measurements within the minute are summarized at the end of that minute. It is **not a forecast of the following minute**, an instantaneous prediction at the beginning of the minute, or a demonstrated route-optimization system.

The target is identified in the registration material as *Vessel Propeller Shaft Mechanical Power*. Its registered QID is:

`2::0::11::0_1::1::0::3::0_14::0::1::0_8`

It occurs in the twelve monthly 2024 datasets but not in the January 2025 file. January 2025 therefore cannot provide labelled evaluation observations for this target.

2. Candidate input measurements

The 2024 records contain **36 observed sensor channels**. Their initial modeling roles are summarized below.

Role	Channels	Proposed treatment
Shaft mechanical power	1	Target variable, subject to quality assessment
Torquemeter-related measurements or close proxy	6	Exclude from model inputs to reduce direct target leakage
Draft measurements with constant zero values	4	Exclude as noninformative measurements
Echosounder water depth	1	Exclude from the initial analysis because October has no paired target intervals
Candidate predictors	**24**	Evaluate for coverage, quality, availability and predictive value

The **24 candidate channels** comprise four engine measurements from other named sources, four motion channels, two onboard relative-wind channels and fourteen environmental provider channels. Relevant examples include engine RPM, turbocharger RPM, scavenging air pressure, fuel load, speed over ground, speed through water, wind and wave conditions, and sea-water current components.

A measurement's presence in the dataset does not by itself establish that it will be available at the intended operational decision time. The engine-source timing and weather-provider issuance times require confirmation. Angular variables such as heading and wind direction also require circular processing rather than naïve arithmetic averaging.

3. Data characteristics affecting the objective

Target reliability. July 2024 contains **28,544 negative shaft-power readings** across fourteen dates, including a near-contiguous sequence of 17,399 readings. November contains **656 negative readings**, with a minimum near **24,757 kW**. These patterns create a substantial risk of unreliable labels. Although the challenge

material warns of torque miscalibration, these readings alone do not establish the precise failure mechanism or fault intervals. The underlying observations should be retained and the affected periods investigated separately.

Observation density. September and October supply **74.9%** and **49.5%**, respectively, of the target readings expected under a nominal uninterrupted 15-second calendar schedule. In October, **88,304 target readings** occupy **22,088 of 44,640 one-minute intervals**, spread over sixteen calendar dates. This is a measure of recorded-data coverage, not proof that the instrument failed when readings were absent.

Environmental measurements. In October, the provider channels have hourly source observations. Although each can be paired with **22,088 target-minute intervals** after past-timestamp alignment, those pairs use only **369 distinct provider timestamps**. Repeated aligned weather values must not be interpreted as independent observations.

Representativeness. Vessel loading, operating regimes, environmental conditions and repeated measurements can change observed relationships. Consequently, the October speed–power association is not evidence of a vessel-wide resistance law.

4. Proposed development and evaluation periods

A chronological evaluation design is necessary because randomly mixing adjacent sensor minutes between training and testing would overstate independence.

Period	Assigned role	Rationale
January–June 2024	Primary training	Earlier labelled observations; quality rules applied before fitting
July 2024	Quality investigation	Sustained negative-power episode
August 2024	Validation	Model selection and parameter decisions on a later period
September–October 2024	Coverage stress analysis	Reduced observation-count density
November 2024	Quality investigation	Extreme negative outliers and further suspicious readings
December 2024	Untouched primary test	Final chronological assessment once methods are fixed

The roles above define a **proposed evaluation**, not completed experiments. The December test period includes **166 negative target readings**. Any quality-filtered evaluation must use rules fixed without looking at test prediction errors, and the number of excluded cases must be reported. July and November remain outside the primary fitting and headline testing periods unless their reliability can be established.

5. Success criteria and expected outputs

Model evaluation will use **mean absolute error (MAE)** and **root mean squared error (RMSE)** in kW, with $\mathbf{R^2}$ reported as a supplementary measure. Results should include observation counts, performance by time block or operating condition, and comparison with appropriately defined simple baselines: a training-mean estimate, a speed-based heuristic fitted only on training data and, when genuinely available, a persistence estimate based on a previously trusted power reading.

The immediate technical criterion is a meaningful improvement over suitable baselines on untouched chronological data. A satisfactory operational error tolerance cannot yet be fixed without a decision context and stakeholder-defined requirements. Claims of fuel savings, route improvements or operational deployment must wait for measured predictive performance and a realistic assessment of input availability.

6. Working assumptions and next decisions

The most consequential open decisions are the reliability boundaries of the torquemeter, the issue-time and timezone meaning of environmental data, the latency of onboard engine measurements, and whether a lagged Noon Report loading indicator would genuinely be available at inference time. These are explicit conditions on the eventual model design, not evidence of achieved model performance.